

Mathematical Properties of the Affine-DDPNM

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1 Preliminaries and notation

We adopt the domain partition, finite-element discretisation, and notation introduced in the accompanying methodology document. The pore domain $\Omega \subset \mathbb{R}^3$ is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$ with m internal interfaces $\{\gamma_k\}_{k=1}^m$. Subdomain Ω_i has m_i ports, of which m_i^u are internal (unknown-traction) and m_i^k are boundary (known-traction).

Interface traction basis. On each interface γ , we introduce a local orthonormal frame $\{\mathbf{n}_\gamma, \mathbf{t}_{\gamma,1}, \mathbf{t}_{\gamma,2}\}$ with origin at the interface centroid $\mathbf{c}_\gamma$, and dimensionless in-plane coordinates $s(\mathbf{x}), t(\mathbf{x}) = O(1)$ on γ . The *mode index set* is

$$\mathcal{M} = \{0, s, t\} \times \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}, \quad (1)$$

with scalar shape functions $\varphi_0 = 1$, $\varphi_s = s$, $\varphi_t = t$ and direction vectors $\mathbf{d}_\mathbf{n} = \mathbf{n}_\gamma$, $\mathbf{d}_{\mathbf{t}_1} = \mathbf{t}_{\gamma,1}$, $\mathbf{d}_{\mathbf{t}_2} = \mathbf{t}_{\gamma,2}$. The traction on γ is expanded as

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n}_\gamma = \sum_{(\alpha, \beta) \in \mathcal{M}} \lambda_{\gamma, \alpha, \beta} \varphi_\alpha(s, t) \mathbf{d}_\beta. \quad (2)$$

For a subdomain Ω_i , let $n_i = 9m_i^u + m_i^k$ be the total number of local traction unknowns (9 per internal interface, 1 per boundary port). Collect them in the vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$. The restriction maps R_i^u, R_i^k and the global unknown vector $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$ are defined analogously to the classic case, with the block size 9×9 per interface instead of 1×1 .

Unit-response loads. For each mode $(\alpha, \beta) \in \mathcal{M}$ on an internal interface $\gamma_{i,r}$ of subdomain Ω_i , the Neumann load vector $\mathbf{f}_{i,r}^{(\alpha, \beta)} \in \mathbb{R}^{n_{i,u}}$ has entries

$$[\mathbf{f}_{i,r}^{(\alpha, \beta)}]_j = - \int_{\gamma_{i,r}} \varphi_\alpha(s, t) \mathbf{d}_\beta \cdot \boldsymbol{\phi}_j^{(i)} \, dS, \quad (3)$$

where $\{\boldsymbol{\phi}_j^{(i)}\}$ is the local velocity basis. For boundary ports only the uniform-normal mode $(\alpha, \beta) = (0, \mathbf{n})$ is used, with a *prescribed* coefficient equal to the inlet/outlet pressure.

Local Stokes system. For each unit load, the discrete Taylor–Hood system on Ω_i reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,r}^{(\alpha, \beta)} \\ \mathbf{X}_{i,r}^{(\alpha, \beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,r}^{(\alpha, \beta)} \\ \mathbf{0} \end{bmatrix}, \quad (4)$$

where $A_i \in \mathbb{R}^{n_{i,u} \times n_{i,u}}$ is the velocity-stiffness matrix (SPD) and $B_i \in \mathbb{R}^{n_{i,p} \times n_{i,u}}$ is the divergence matrix. The pair $(V_{i,h}, Q_{i,h})$ is the inf–sup stable Taylor–Hood $[P_2]^3$ – P_1 element.

For a boundary port $\gamma_{i,r} \in \mathcal{K}_i$, the single $(\alpha, \beta) = (0, \mathbf{n})$ load is used.

Local DtN operator. The local Dirichlet-to-Neumann (traction-to-flux) map $\mathbf{G}_i \in \mathbb{R}^{n_i \times n_i}$ has entries

$$\mathbf{G}_i[(\gamma, \alpha, \beta), (\gamma', \alpha', \beta')] = -[\mathbf{f}_i^{(\alpha, \beta)}(\gamma)]^\top \mathbf{U}_i^{(\alpha', \beta')}(\gamma'), \quad (5)$$

where (γ, α, β) ranges over the n_i local modes. For boundary ports, only the block $(\gamma, 0, \mathbf{n})$ is stored. $\mathbf{G}_i$ is block-partitioned into $\mathbf{G}_i^{uu}$ (unknown–unknown) and $\mathbf{G}_i^{uk}$ (unknown–known) as in the classic case.

Global Schur complement. Enforcing flux continuity on every internal interface (constant-mode normal component only) yields the global system

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k, \quad (6)$$

where $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$ collects the prescribed inlet/outlet tractions and $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$.

2 Mathematical properties

We now establish the solvability, symmetry, positive-definiteness, and mass-conservation properties of the Affine-DDPNM formulation. The overall structure parallels the classic DD-PNM analysis [1], with careful treatment of the nine-mode interface space.

Lemma 2.1 (Local well-posedness). *For every interface-traction vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$, the discrete Stokes subproblem in each Ω_i admits a unique solution $(\mathbf{U}_i, \mathbf{X}_i)$.*

Proof. The saddle-point matrix $K_i = \begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix}$ is invertible because (i) A_i is SPD and (ii) the Taylor–Hood pair satisfies the discrete inf–sup condition on the body-fitted tetrahedral mesh of Ω_i . The right-hand side of (4) is a linear combination of the primitive load vectors $\mathbf{f}_{i,r}^{(\alpha,\beta)}$ with coefficients $\boldsymbol{\lambda}_i$; hence a unique solution exists for any $\boldsymbol{\lambda}_i$. $\square$

Lemma 2.2 (Discrete incompressibility of the unit responses). *For each internal interface $\gamma_{i,r}$ and each mode $(\alpha, \beta) \in \mathcal{M}$, the velocity response $\mathbf{U}_{i,r}^{(\alpha,\beta)}$ is discretely divergence-free:*

$$B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}. \quad (7)$$

Proof. Equation (7) follows directly from the second block row of the local system (4). For completeness, we also provide the explicit expression of the response. From the first block row,

$$\mathbf{U}_{i,r}^{(\alpha,\beta)} = A_i^{-1} \left(\mathbf{f}_{i,r}^{(\alpha,\beta)} - B_i^\top \mathbf{X}_{i,r}^{(\alpha,\beta)} \right).$$

Imposing $B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}$ yields

$$\mathbf{X}_{i,r}^{(\alpha,\beta)} = M_i^{-1} B_i A_i^{-1} \mathbf{f}_{i,r}^{(\alpha,\beta)}, \quad M_i := B_i A_i^{-1} B_i^\top \text{ (SPD)}.$$

Substituting back,

$$\mathbf{U}_{i,r}^{(\alpha,\beta)} = \underbrace{(A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1})}_{=: H_i} \mathbf{f}_{i,r}^{(\alpha,\beta)}, \quad (8)$$

where H_i is the velocity–velocity block of K_i^{-1} . Since $B_i H_i = 0$ (the operator H_i maps any load vector into $\ker B_i$), we have $B_i \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}$. $\square$

Lemma 2.3 (Symmetry and energy identity for the local DtN map). *Let $\mathbf{G}_i$ be defined by (5). Then $\mathbf{G}_i$ is symmetric and, for any local traction vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$,*

$$\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = -\mathbf{U}_i^\top A_i \mathbf{U}_i \leq 0, \quad (9)$$

where $\mathbf{U}_i = \sum_{\gamma,\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta} \mathbf{U}_i^{(\alpha,\beta)}(\gamma)$ is the velocity field induced by $\boldsymbol{\lambda}_i$. In particular, $\mathbf{G}_i$ is symmetric negative semi-definite.

Proof. Denote the two index pairs by $\mu = (\gamma, \alpha, \beta)$ and $\mu' = (\gamma', \alpha', \beta')$. From Lemma 2.2, $B_i \mathbf{U}_i^{(\mu')} = \mathbf{0}$. Using the explicit form (8), $\mathbf{U}_i^{(\mu)} = H_i \mathbf{f}_i^{(\mu)}$ where

$$H_i := A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}, \quad M_i := B_i A_i^{-1} B_i^\top. \quad (10)$$

We verify the identity $H_i^\top A_i H_i = H_i$ by direct computation:

$$\begin{aligned} H_i^\top A_i H_i &= (A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}) A_i (A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}) \\ &= A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} \\ &\quad + A_i^{-1} B_i^\top M_i^{-1} (B_i A_i^{-1} B_i^\top) M_i^{-1} B_i A_i^{-1} \\ &= A_i^{-1} - 2A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} + A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} \\ &= H_i, \end{aligned}$$

where in the penultimate line the parenthesised term equals M_i , cancelling one M_i^{-1} and reducing the coefficient from 2 to 1. A direct consequence is $B_i H_i = 0$ (apply B_i to both sides of $H_i^\top A_i H_i = H_i$ and use $M_i = B_i A_i^{-1} B_i^\top$).

Now compute the DtN entry:

$$\begin{aligned} \mathbf{G}_i(\mu, \mu') &= -[\mathbf{f}_i^{(\mu)}]^\top \mathbf{U}_i^{(\mu')} = -[\mathbf{f}_i^{(\mu)}]^\top H_i \mathbf{f}_i^{(\mu')} \\ &= -[\mathbf{f}_i^{(\mu)}]^\top (H_i^\top A_i H_i) \mathbf{f}_i^{(\mu')} = -[H_i \mathbf{f}_i^{(\mu)}]^\top A_i [H_i \mathbf{f}_i^{(\mu')}] \\ &= -[\mathbf{U}_i^{(\mu)}]^\top A_i \mathbf{U}_i^{(\mu')}. \end{aligned}$$

The final expression is symmetric in (μ, μ') by the symmetry of A_i , hence $\mathbf{G}_i$ is symmetric. For the energy identity, expand

$$\begin{aligned} \boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i &= - \sum_{\mu, \mu'} \lambda_{i, \mu} \lambda_{i, \mu'} [\mathbf{U}_i^{(\mu)}]^\top A_i \mathbf{U}_i^{(\mu')} \\ &= - \left(\sum_{\mu} \lambda_{i, \mu} \mathbf{U}_i^{(\mu)} \right)^\top A_i \left(\sum_{\mu'} \lambda_{i, \mu'} \mathbf{U}_i^{(\mu')} \right) \\ &= -\mathbf{U}_i^\top A_i \mathbf{U}_i \leq 0, \end{aligned}$$

where the non-positivity follows because A_i is SPD. $\square$

Lemma 2.4 (Kernel of the local DtN map). *Let $\mathbf{1}_{(0, \mathbf{n})} \in \mathbb{R}^{n_i}$ be the vector with entry 1 at every internal-interface uniform-normal mode $(\gamma, 0, \mathbf{n})$ and entry 0 elsewhere; and with entry 1 at every boundary-port (single) mode. Then*

(i) $\mathbf{G}_i \mathbf{1}_{(0, \mathbf{n})} = \mathbf{0}$.

(ii) $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$ if and only if $\boldsymbol{\lambda}_i \in \text{span}\{\mathbf{1}_{(0, \mathbf{n})}\}$.

Consequently, $\ker(\mathbf{G}_i) = \text{span}\{\mathbf{1}_{(0, \mathbf{n})}\}$ is exactly one-dimensional.

Proof. (i) Set all uniform-normal coefficients to the same value p_{const} and all other coefficients to zero. The right-hand side of the local Stokes system (4) becomes $\sum_{\gamma} p_{\text{const}} \mathbf{f}_i^{(\gamma, 0, \mathbf{n})}$. By the divergence theorem and the definition $\mathbf{f}_i^{(\gamma, 0, \mathbf{n})} = - \int_{\gamma, \gamma} \boldsymbol{\phi}_j^{(i)} \cdot \mathbf{n}_{i, \gamma} dS$,

$$\sum_{\gamma} \mathbf{f}_i^{(\gamma, 0, \mathbf{n})} = - \int_{\partial \Omega_i} \boldsymbol{\phi}_j^{(i)} \cdot \mathbf{n} dS = - \int_{\Omega_i} \nabla \cdot \boldsymbol{\phi}_j^{(i)} d\mathbf{x} = [B_i^\top \mathbf{1}]_j,$$

where $\mathbf{1}$ is the coefficient vector of the constant pressure $p_h \equiv 1$ in $Q_{i, h}$. Hence $(\mathbf{U}_i, \mathbf{X}_i) = (\mathbf{0}, p_{\text{const}} \mathbf{1})$ solves the local system, and by uniqueness (Lemma 2.1), $\mathbf{U}_i = \mathbf{0}$. The energy identity (9) then gives $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$. Since $\mathbf{G}_i$ is symmetric negative semi-definite, this implies $\mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{0}$ as well.

(ii) ($\Leftarrow$) follows from (i).

($\Rightarrow$) Suppose $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$. By the energy identity, $\mathbf{U}_i^\top A_i \mathbf{U}_i = 0$, and since A_i is SPD we obtain $\mathbf{U}_i = \mathbf{0}$.

The discrete velocity $\mathbf{U}_i = \mathbf{0}$ implies that the corresponding FE velocity field $\mathbf{u}_{i,h} \equiv \mathbf{0}$ in Ω_i . Substituting $\mathbf{u}_{i,h} = \mathbf{0}$ into the weak Stokes formulation (equivalently, into the first block row of (4)), the momentum equation reduces to

$$\int_{\Omega_i} p_{i,h} (\nabla \cdot \mathbf{v}_h) \, d\mathbf{x} = \sum_{\gamma} \int_{\gamma_{i,\gamma}} \mathbf{t}_{i,\gamma}(\mathbf{x}) \cdot \mathbf{v}_h \, dS \quad \forall \mathbf{v}_h \in V_{i,h}, \quad (11)$$

where $\mathbf{t}_{i,\gamma}$ is the applied traction on $\gamma_{i,\gamma}$ given by the affine Ansatz (2).

Choosing test functions $\mathbf{v}_h$ supported in the interior of Ω_i (where the boundary integral vanishes) yields $\int_{\Omega_i} p_{i,h} (\nabla \cdot \mathbf{v}_h) \, d\mathbf{x} = 0$ for all discretely divergence-free interior test functions. Since the Taylor–Hood pair is inf-sup stable, this forces $\nabla p_{i,h} = \mathbf{0}$ in Ω_i ; hence $p_{i,h} \equiv p_0$ (a constant) throughout the pore.

With $p_{i,h} \equiv p_0$, the left-hand side of (11) becomes $p_0 \int_{\Omega_i} \nabla \cdot \mathbf{v}_h \, d\mathbf{x} = p_0 \int_{\partial\Omega_i} \mathbf{v}_h \cdot \mathbf{n} \, dS$. Equation (11) therefore reads

$$\sum_{\gamma} \int_{\gamma_{i,\gamma}} (\mathbf{t}_{i,\gamma}(\mathbf{x}) - p_0 \mathbf{n}_{i,\gamma}) \cdot \mathbf{v}_h \, dS = 0 \quad \forall \mathbf{v}_h \in V_{i,h}.$$

Since the trace of $V_{i,h}$ is dense on each interface independently, this forces $\mathbf{t}_{i,\gamma}(\mathbf{x}) \equiv p_0 \mathbf{n}_{i,\gamma}$ pointwise on each $\gamma_{i,\gamma}$.

Inserting the affine expansion (2):

$$\sum_{\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta} \varphi_{\alpha}(s, t) \mathbf{d}_{\beta} \equiv p_0 \mathbf{n}_{i,\gamma} \quad \text{on each } \gamma_{i,\gamma}.$$

The right-hand side $p_0 \mathbf{n}_{i,\gamma}$ is a constant multiple of the normal vector. Linear independence of the nine basis functions $\{\varphi_{\alpha} \mathbf{d}_{\beta}\}$ on $\gamma_{i,\gamma}$ then forces

$$\lambda_{i,\gamma,0,\mathbf{n}} = p_0 \quad (\text{same for all } \gamma), \quad \lambda_{i,\gamma,\alpha,\beta} = 0 \quad \forall (\alpha, \beta) \neq (0, \mathbf{n}).$$

Therefore $\boldsymbol{\lambda}_i \in \text{span}\{\mathbf{1}_{(0,\mathbf{n})}\}$.

The one-dimensionality follows since $\mathbf{1}_{(0,\mathbf{n})} \neq \mathbf{0}$. $\square$

Remark 2.5. Lemma 2.4 has a clear physical interpretation: the only way to obtain zero velocity in a pore subdomain (which has at least one no-slip solid wall, fixing rigid-body modes) is to apply a uniform constant pressure on all its boundaries. Any spatial variation of the normal traction (the $\mathbf{s}\mathbf{n}$, $\mathbf{t}\mathbf{n}$ modes) or any tangential traction component drives a non-trivial Stokes flow. The kernel therefore remains one-dimensional despite the nine-fold enrichment of the interface space.

Theorem 2.6 (Solvability and mass conservation of Affine-DDPNM). *Let $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$ be the global Schur complement defined in (6) and let $\mathbf{F}$ be the corresponding forcing vector. Then:*

- (i) **Positive definiteness.** $\mathbf{S}$ is symmetric positive-definite (SPD).
- (ii) **Global solvability.** The linear system $\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}$ has a unique solution $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$.
- (iii) **Mass conservation.** The reconstructed velocity field $\mathbf{u}_h^{DD} = \sum_i \mathbf{u}_{i,h}$ satisfies both local and global discrete mass balances:

$$\sum_{r=1}^{m_i} Q_{i,r}^{(0,\mathbf{n})} = 0 \quad (\text{local, each } \Omega_i), \quad \int_{\partial\Omega} \mathbf{u}_h^{DD} \cdot \mathbf{n} \, dS = 0 \quad (\text{global}). \quad (12)$$

Proof. (i) **Symmetry and positive definiteness.** Symmetry of $\mathbf{S}$ follows from the symmetry of each $\mathbf{G}_i^{uu}$ (Lemma 2.3) and the Boolean nature of the restriction matrices R_i^u .

For any $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, let $\boldsymbol{\lambda}_i^u = R_i^u \boldsymbol{\lambda}$ be the local unknown-traction vector on Ω_i (with zero entries for boundary ports). By the energy identity (9),

$$\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = - \sum_{i=1}^N (\boldsymbol{\lambda}_i^u)^\top \mathbf{G}_i^{uu} \boldsymbol{\lambda}_i^u = \sum_{i=1}^N \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0.$$

If $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = 0$, then $\mathbf{U}_i = \mathbf{0}$ for every i . By Lemma 2.4(ii), each $\boldsymbol{\lambda}_i^u$ must belong to $\text{span}\{\mathbf{1}_{(0,\mathbf{n})}\}$ on the internal interfaces of Ω_i .

Global consistency across adjacent pores forces all uniform-normal coefficients to share the same value $c \in \mathbb{R}$ on every internal interface. However, the boundary ports of each pore carry *prescribed* (not unknown) tractions. The only constant value compatible with $\boldsymbol{\lambda}_i^u$ having zero entries on boundary ports (since $\boldsymbol{\lambda}_i^u$ is formed by restriction of the purely internal vector $\boldsymbol{\lambda}$) is $c = 0$. Hence $\boldsymbol{\lambda} = \mathbf{0}$, and $\mathbf{S}$ is SPD.

(ii) Global solvability. Immediate from (i): an SPD matrix is invertible.

(iii) Mass conservation. For each subdomain Ω_i , choose the constant pressure test function $q_h \equiv 1 \in Q_{i,h}$ in the discrete incompressibility constraint $B_i \mathbf{U}_i = \mathbf{0}$. By the divergence theorem and the no-slip condition on solid walls,

$$0 = \int_{\Omega_i} \nabla \cdot \mathbf{u}_{i,h} \, d\mathbf{x} = \sum_{r=1}^{m_i} \int_{\gamma_{i,r}} \mathbf{u}_{i,h} \cdot \mathbf{n}_{i,r} \, dS = \sum_{r=1}^{m_i} Q_{i,r}^{(0,\mathbf{n})},$$

which establishes the local mass balance.

The global Schur assembly (6) enforces $Q_{i,r}^{(0,\mathbf{n})} + Q_{j,r}^{(0,\mathbf{n})} = 0$ on every internal interface shared by pores i and j . Summing the local balances over all pores, the internal flux pairs cancel pairwise and only the inlet/outlet contributions remain, yielding $\int_{\partial\Omega} \mathbf{u}_h^{\text{DD}} \cdot \mathbf{n} \, dS = 0$. $\square$

3 Error estimate as a Galerkin projection

Following Appendix B of [1], we formulate the classical non-overlapping domain-decomposition (C-DD) discrete Stokes system and characterise the Affine-DDPNM as its Galerkin restriction onto the nine-mode affine interface space. We then prove the equivalence between the Galerkin construction and the assembled DtN-based Schur system (6), and obtain the error estimate as a direct consequence.

3.1 C-DD formulation

We recall the C-DD formulation of Appendix B.1 in [1], extended to three dimensions. On each internal interface $\gamma_{i,r}$ of subdomain Ω_i , the traction is expanded in the FE face basis $\{\psi_\ell\}$ (the restriction of the velocity trace basis to the face):

$$\mathbf{t}_{i,r}^{\text{FE}}(\mathbf{x}) = \sum_{\ell=1}^{m_{i,r}} \left(\eta_{i,r}^\ell \mathbf{n}_{i,r} + \tau_{i,r}^{\ell,1} \mathbf{t}_{i,r}^1 + \tau_{i,r}^{\ell,2} \mathbf{t}_{i,r}^2 \right) \psi_\ell(\mathbf{x}), \quad (13)$$

where $m_{i,r}$ is the number of velocity DOFs on $\gamma_{i,r}$, and $\eta_{i,r}^\ell, \tau_{i,r}^{\ell,1}, \tau_{i,r}^{\ell,2} \in \mathbb{R}$ are the normal and two tangential traction coefficients at node ℓ . Let $\boldsymbol{\eta}_{i,r} \in \mathbb{R}^{m_{i,r}}$ and $\boldsymbol{\tau}_{i,r}^1, \boldsymbol{\tau}_{i,r}^2 \in \mathbb{R}^{m_{i,r}}$ collect the nodal values. The *full* C-DD traction vector on $\gamma_{i,r}$ is $\boldsymbol{\xi}_{i,r} := (\boldsymbol{\eta}_{i,r}; \boldsymbol{\tau}_{i,r}^1; \boldsymbol{\tau}_{i,r}^2) \in \mathbb{R}^{3m_{i,r}}$.

The C-DD Neumann coupling blocks on $\gamma_{i,r}$ (cf. Appendix B.1 of [1]) are

$$N_{i,r}^{\mathbf{n}} \in \mathbb{R}^{m_{i,r} \times n_{i,u}}, \quad N_{i,r}^{\mathbf{t}^1} \in \mathbb{R}^{m_{i,r} \times n_{i,u}}, \quad N_{i,r}^{\mathbf{t}^2} \in \mathbb{R}^{m_{i,r} \times n_{i,u}},$$

with entries $[N_{i,r}^{\mathbf{n}}]_{\ell,j} = \int_{\gamma_{i,r}} \psi_\ell(\phi_j^{(i)} \cdot \mathbf{n}_{i,r}) \, dS$ and analogously for the tangential blocks. Stacking all interfaces of Ω_i , the local C-DD coupling matrix is $N_i = [N_i^{\mathbf{n}} \ N_i^{\mathbf{t}^1} \ N_i^{\mathbf{t}^2}]^\top$, and the local traction vector is $\boldsymbol{\xi}_i = (\boldsymbol{\eta}_i; \boldsymbol{\tau}_i^1; \boldsymbol{\tau}_i^2)$.

The discrete Stokes system on Ω_i in C-DD form reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} V_i \\ P_i \end{bmatrix} = - \begin{bmatrix} N_i^\top \boldsymbol{\xi}_i \\ \mathbf{0} \end{bmatrix} + (\text{inlet/outlet contributions}).$$

Eliminating the interior velocity and pressure yields the C-DD Schur complement $\widehat{\mathbf{S}}_i$ on Ω_i :

$$\widehat{\mathbf{S}}_i = N_i H_i N_i^\top \in \mathbb{R}^{(3 \sum_r m_{i,r}) \times (3 \sum_r m_{i,r})},$$

where H_i is the velocity–velocity block of K_i^{-1} defined in (10). Global assembly with Boolean restriction matrices $\widehat{R}_i^u$ (mapping local C-DD face DOFs to global interface DOFs) gives the C-DD system

$$\widehat{\mathbf{S}} \boldsymbol{\xi}^{\text{FE}} = \widehat{\mathbf{F}}, \quad \widehat{\mathbf{S}} = \sum_{i=1}^N (\widehat{R}_i^u)^\top \widehat{\mathbf{S}}_i \widehat{R}_i^u, \quad (14)$$

with $\widehat{\mathbf{S}}$ SPD and $\boldsymbol{\xi}^{\text{FE}} \in \mathbb{R}^{m_\Gamma}$ collecting all global interface nodal traction DOFs (both normal and tangential).

3.2 Affine restriction operator and equivalence

For each interface γ_k , let $M_k \in \mathbb{R}^{m_k \times m_k}$ be the face mass matrix, $(M_k)_{\ell\ell'} = \int_{\gamma_k} \psi_\ell \psi_{\ell'} \, dS$. For each of the nine affine modes $(\alpha, \beta) \in \mathcal{M}$, define the right-hand side vector $\mathbf{b}_k^{\alpha, \beta, \gamma} \in \mathbb{R}^{m_k}$ and assemble the local restriction matrix $P_k^{\text{aff}} \in \mathbb{R}^{3m_k \times 9}$ as

$$P_k^{\text{aff}} = \begin{bmatrix} P_k^{0, \mathbf{n}} & P_k^{0, \mathbf{t}_1} & P_k^{0, \mathbf{t}_2} & P_k^{s, \mathbf{n}} & P_k^{s, \mathbf{t}_1} & P_k^{s, \mathbf{t}_2} & P_k^{t, \mathbf{n}} & P_k^{t, \mathbf{t}_1} & P_k^{t, \mathbf{t}_2} \end{bmatrix}, \quad (15)$$

where each column is the L^2 -projection of the corresponding affine mode onto the FE face basis:

$$M_k P_k^{\alpha, \mathbf{n}} = \mathbf{b}_k^{\alpha, \mathbf{n}}, \quad (\mathbf{b}_k^{\alpha, \mathbf{n}})_\ell = \int_{\gamma_k} \varphi_\alpha(s, t) \psi_\ell(\mathbf{x}) \, dS,$$

and analogously $M_k P_k^{\alpha, \mathbf{t}_\nu} = \mathbf{b}_k^{\alpha, \mathbf{t}_\nu}$ with $\mathbf{t}_\nu$ instead of $\mathbf{n}_k$ in the integrand. The three component blocks (normal, tangent-1, tangent-2) are stacked vertically in (15).

The *global* affine restriction operator is the block-diagonal assembly

$$\mathcal{P}_{\text{aff}} = \bigoplus_{k=1}^m P_k^{\text{aff}} : \mathbb{R}^{9m} \rightarrow \mathbb{R}^{m_\Gamma}, \quad (16)$$

mapping the $9m$ affine coefficients to the m_Γ C-DD nodal traction DOFs.

Lemma 3.1 (Equivalence of the assembled and projected Schur systems). *Let $\mathbf{S}$ be the DtN-assembled Schur complement (6) and let $\mathbf{F}$ be the corresponding forcing vector. Let $\mathcal{P}_{\text{aff}}$ be defined by (16). Then*

$$\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}} \mathcal{P}_{\text{aff}}, \quad \mathbf{F} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{F}}, \quad (17)$$

where $\widehat{\mathbf{S}}$ and $\widehat{\mathbf{F}}$ are the C-DD system matrices (14).

Proof. Consider a single pore Ω_i . For each internal interface $\gamma_{i,r}$ of Ω_i , the C-DD coupling block $N_{i,r}^{\mathbf{n}}$ acts on the nodal normal traction $\boldsymbol{\eta}_{i,r}$. The restriction matrix $P_{i,r}^{\text{aff}}$ maps the nine affine coefficients $\boldsymbol{\lambda}_{i,r} \in \mathbb{R}^9$ to $(\boldsymbol{\eta}_{i,r}; \boldsymbol{\tau}_{i,r}^1; \boldsymbol{\tau}_{i,r}^2)$. The affine primitive load $\mathbf{f}_i^{(\gamma, \alpha, \beta)}$ defined in (3) is exactly the column of $N_i^\top$ corresponding to the L^2 -projected mode $P_k^{\alpha, \beta}$:

$$\mathbf{f}_i^{(\gamma, \alpha, \beta)} = -(N_{i,r}^{\mathbf{n}})^\top P_{i,r}^{\alpha, \mathbf{n}} - (N_{i,r}^{\mathbf{t}_1})^\top P_{i,r}^{\alpha, \mathbf{t}_1} - (N_{i,r}^{\mathbf{t}_2})^\top P_{i,r}^{\alpha, \mathbf{t}_2}.$$

Consequently, assembling the m_i interfaces, the matrix of all primitive load vectors for Ω_i equals $N_i^\top P_i^{\text{aff}}$, where P_i^{aff} is the block-diagonal assembly of the per-interface restriction matrices. The local DtN block (5) is then

$$\mathbf{G}_i = -(N_i^\top P_i^{\text{aff}})^\top H_i (N_i^\top P_i^{\text{aff}}) = -(P_i^{\text{aff}})^\top N_i H_i N_i^\top P_i^{\text{aff}} = -(P_i^{\text{aff}})^\top \hat{\mathbf{S}}_i P_i^{\text{aff}}.$$

The global Schur complement (6) assembles $\mathbf{S} = -\sum_i (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u$. Incorporating the Boolean maps R_i^u and $\hat{R}_i^u$, $\mathbf{S}$ and $\mathcal{P}_{\text{aff}}^\top \hat{\mathbf{S}} \mathcal{P}_{\text{aff}}$ coincide block-wise on every interface. The same computation for the forcing term yields $\mathbf{F} = \mathcal{P}_{\text{aff}}^\top \hat{\mathbf{F}}$. $\square$

3.3 Error estimate

Theorem 3.2 (Error estimate for Affine-DDPNM). *Let $\boldsymbol{\xi}^{FE}$ be the C-DD interface traction solving (14). Let $\boldsymbol{\lambda}^{\text{aff}}$ be the Affine-DDPNM solution ($\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} = \mathbf{F}$) and set $\boldsymbol{\xi}^{\text{aff}} = \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}^{\text{aff}} \in \mathbb{R}^{m_\Gamma}$. Then:*

(i) **Optimality.**

$$\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}\|_{\hat{\mathbf{S}}} = \min_{\boldsymbol{\mu} \in \mathbb{R}^{9m}} \|\boldsymbol{\xi}^{FE} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}. \quad (18)$$

(ii) **Velocity energy equivalence.**

$$\sum_{i=1}^N \|\mathbf{U}_i^{FE} - \mathbf{U}_i^{\text{aff}}\|_{A_i}^2 = \|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}\|_{\hat{\mathbf{S}}}^2. \quad (19)$$

Proof. By Lemma 3.1, the Affine-DDPNM system is exactly the Galerkin restriction of the C-DD system onto $\text{Range}(\mathcal{P}_{\text{aff}})$. The Galerkin orthogonality $\mathcal{P}_{\text{aff}}^\top \hat{\mathbf{S}} (\boldsymbol{\xi}^{\text{aff}} - \boldsymbol{\xi}^{FE}) = \mathbf{0}$ follows by subtracting $\mathcal{P}_{\text{aff}}^\top (\hat{\mathbf{S}} \boldsymbol{\xi}^{FE} - \hat{\mathbf{F}}) = \mathbf{0}$ from $\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} - \mathbf{F} = \mathbf{0}$ and applying (17).

Hence the error $\mathbf{e} := \boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}$ is $\hat{\mathbf{S}}$ -orthogonal to $\text{Range}(\mathcal{P}_{\text{aff}})$. For any $\boldsymbol{\mu} \in \mathbb{R}^{9m}$, C  a's lemma in the $\hat{\mathbf{S}}$ -inner product gives

$$\|\boldsymbol{\xi}^{FE} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}^2 = \|\mathbf{e}\|_{\hat{\mathbf{S}}}^2 + \|\boldsymbol{\xi}^{\text{aff}} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}^2 \geq \|\mathbf{e}\|_{\hat{\mathbf{S}}}^2,$$

with equality when $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ (since $\mathcal{P}_{\text{aff}}$ is injective on the space of solutions with zero boundary data). This proves (18). Equation (19) follows from the C-DD energy identity (Appendix B.1 of [1]) applied to $\boldsymbol{\xi}^{FE}$ and $\boldsymbol{\xi}^{\text{aff}}$. $\square$

3.4 Discussion

Remark 3.3 (Enrichment guarantee). *The classic DD-PNM restriction operator $\mathcal{P}_{P0}$ maps $\mathbb{R}^m$ to $\mathbb{R}^{m_\Gamma}$ by L^2 -projecting the constant normal function onto the FE face basis (in the normal direction only). Because the constant normal mode $(\alpha, \beta) = (0, \mathbf{n})$ is one of the nine affine modes, $\text{Range}(\mathcal{P}_{P0}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$. Theorem 3.2 therefore guarantees*

$$\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}\|_{\hat{\mathbf{S}}} \leq \|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{P0}\|_{\hat{\mathbf{S}}},$$

i.e. the Affine-DDPNM error in the C-DD energy norm is never larger than the classic DD-PNM error. This provides a rigorous justification for the 73–90% error reduction observed numerically.

Remark 3.4 (Mesh dependence of the projection error). *The optimality result (18) bounds the error by the best-approximation error of the discrete C-DD traction $\boldsymbol{\xi}^{FE}$ in the affine subspace $\text{Range}(\mathcal{P}_{\text{aff}})$, measured in the mesh-dependent C-DD energy norm $\|\cdot\|_{\hat{\mathbf{S}}}$. This distance has two components:*

1. The **projection error** of the true continuum interface traction onto the affine subspace. This is determined by the interface basis alone and is h -independent.
2. The **FE discretisation error** of the C-DD traction, which vanishes as $h \rightarrow 0$.

As the mesh is refined, $\widehat{\mathbf{S}}$ converges to the continuous Steklov–Poincaré operator, and $\|\boldsymbol{\xi}^{FE} - \mathcal{P}_{\text{aff}}\boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}$ converges to the continuous best-approximation error. This limiting error is generically non-zero; mesh refinement alone cannot drive the DDPNM error to zero—further enrichment of the interface space (e.g. quadratic shape functions $\{s^2, st, t^2\}$) is required. This explains the numerical observation that the Affine-DDPNM velocity L^2 error is essentially mesh-independent within the tested range (4.3K–45K tetrahedra).

References

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